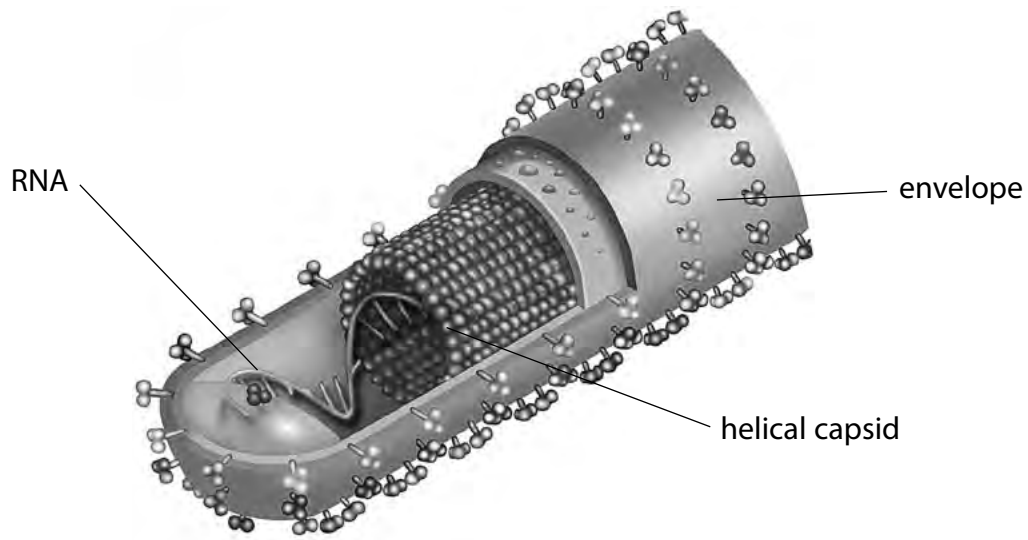


- 1 (a) The diagram shows the structure of an Ebola virus.



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- (i) Which of the following viruses contain the same type of nucleic acid as the Ebola virus?

(1)

- ☐ A human immunodeficiency virus (HIV) only
- ☐ B human immunodeficiency virus (HIV) and tobacco mosaic virus (TMV)
- ☐ C lambda phage ( $\lambda$  phage) only
- ☐ D lambda phage ( $\lambda$  phage) and tobacco mosaic virus (TMV)

- (ii) Which of the following viruses have a helical capsid?

(1)

- ☐ A human immunodeficiency virus (HIV) and lambda phage ( $\lambda$  phage)
- ☐ B human immunodeficiency virus (HIV) only
- ☐ C lambda phage ( $\lambda$  phage) and tobacco mosaic virus (TMV)
- ☐ D tobacco mosaic virus (TMV) only

(iii) What type of molecule makes up the capsid of a virus?

(1)

- ☐ **A** carbohydrate
- ☐ **B** lipid
- ☐ **C** nucleic acid
- ☐ **D** protein

(iv) The volume of an Ebola virus is approximately  $7.76 \times 10^4 \text{ nm}^3$ .

Tobacco mosaic virus (TMV) is approximately 300 nm long and 80 nm in diameter.

Calculate how many times larger Ebola virus is than TMV.

Assume that TMV is a cylinder in shape.

The volume of a cylinder is calculated using the formula

$$V = \pi r^2 l$$

(2)

Answer.....

(b) Human immunodeficiency virus (HIV) contains two enzymes that are **not** found in most other types of virus.

(i) Name these two enzymes.

(1)

(ii) Explain why HIV contains these two enzymes.

(3)

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(Total for Question 1 = 9 marks)